

```
print(v1 == v3)
print(v1 < v2)
print(v1 > v2)
print(abs(v1))
print(abs(v2))
print()

# These lines print Vectors (calls the __str__() method)
print('Vector 1:', v1)
print('Vector 2:', v2)
print('Vector 1 + Vector 2:', v1 + v2)
print('Vector 1 - Vector 2:', v1 - v2)
print('Vector 1 times Vector 2:', v1 * v2)
print('Vector 2 times 5:', v1 * 5)
```

Listing 9-9: Sample main code that creates and compares `Vector`s, does math, and prints `Vector`s

This generates the following output:

```
False
True
False
True
5.0
2.8284271247461903

Vector 1: This vector has the value (3, 4)
Vector 2: This vector has the value (2, 2)
Vector 1 + Vector 2: This vector has the value (5, 6)
Vector 1 - Vector 2: This vector has the value (1, 2)
Vector 1 times Vector 2: This vector has the value (6, 8)
Vector 2 times 5: This vector has the value (15, 20)
```

The first set of calls to `print()` output Boolean and numeric values, which result from calling math and comparison operator magic methods. In the second set, we print two `Vector` objects, then compute and print some new `Vectors`. Internally, the `print()` function first calls Python's `str()` function for each item to be printed; that results in a call to the `Vector`'s `__str__()` magic method, which creates a formatted string with the relevant information.